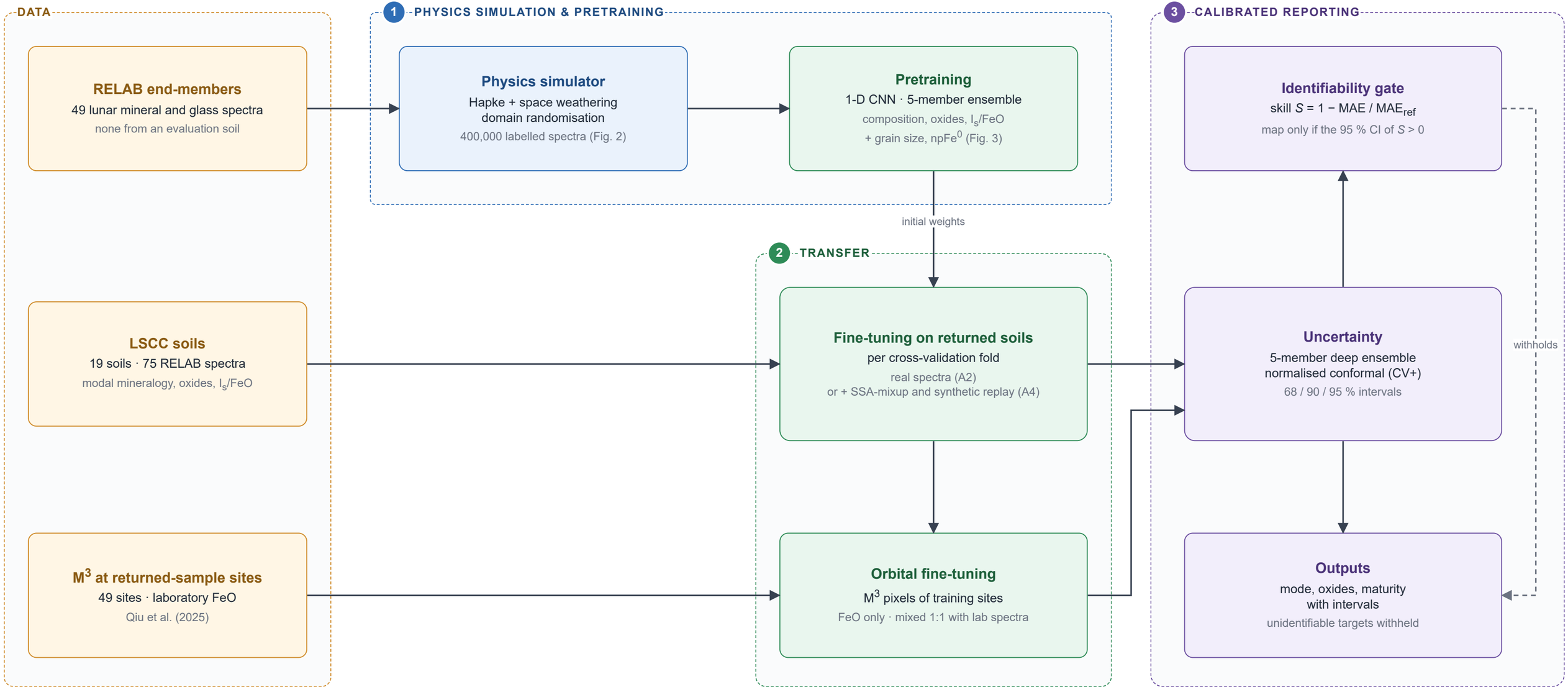

DATA
RELAB end-members
49 lunar mineral and glass spectra
none from an evaluation soil
LSCC soils
19 soils · 75 RELAB spectra
modal mineralogy, oxides, I_s/FeO
M³ at returned-sample sites
49 sites · laboratory FeO
Qiu et al. (2025)
1 PHYSICS SIMULATION & PRETRAINING
Physics simulator
Hapke + space weathering
domain randomisation
400,000 labelled spectra (Fig. 2)
Pretraining
1-D CNN · 5-member ensemble
composition, oxides, I_s/FeO
+ grain size, npFe⁰ (Fig. 3)
initial weights
2 TRANSFER
Fine-tuning on returned soils
per cross-validation fold
real spectra (A2)
or + SSA-mixup and synthetic replay (A4)
Orbital fine-tuning
M³ pixels of training sites
FeO only · mixed 1:1 with lab spectra
3 CALIBRATED REPORTING
Identifiability gate
skill S = 1 − MAE / MAE_ref
map only if the 95 % CI of S > 0
Uncertainty
5-member deep ensemble
normalised conformal (CV+)
68 / 90 / 95 % intervals
Outputs
mode, oxides, maturity
with intervals
unidentifiable targets withheld
withholds